

SEP (Separator)

Overview

The Separator (SEP) cabinet is a required when using a chiller which provides chilled water at temperatures between 6 and 12°C.

The SEP fulfills two functions:

- heat exchange of 6-12°C primary to 20°C secondary
- provide separation and protection of MR System from chiller

Specifications

Specification	Value
Primary	
min. flow in primary circuit	90 l/min.
temperature bandwidth	6-12°C
maximal pressure difference	1.5 bar
Secondary	
nominal temp	20°C
temperature bandwidth (temp may leave this bandwidth for maximum 30 sec.)	19-22°C
alarm temp	≤15 or ≥26
max pressure difference	3.5 bar
Flow	140 l/min.

Water Circuit

Function

In case of a power failure the SEP must turn itself on again automatically.

Two **pressure gauges** placed at pump intake (5a) and pump output (5b) are supplied for refilling and troubleshooting.

The static fill pressure (= pump off) is approximately 0.5 - 1.0 bar.

Older systems with a built-in low-pressure switch require a slightly higher fill pressure of 1.5 - 2.0 bar. For the dynamic pressure (= pump on), the pressure difference between both gauges should read 3.5 bar + 0 / - 0.5 bar. Example: Pin = 0.5 bar, therefore Pout must be 3.5 - 4.0 bar (dynamic).

A **safety valve** (16) with a water run-off hose at pump intake is set for 3 bar over-pressure. This prevents the SEP from over-filling.

The **bypass valve** (11) adjusts the pressure difference to 3.5 bar + 0 / - 0.5 bar and prevents over-pressure on MR system components.

The regulation maintains the water temp ≤2K/5min with a running integrations time of 30 s.

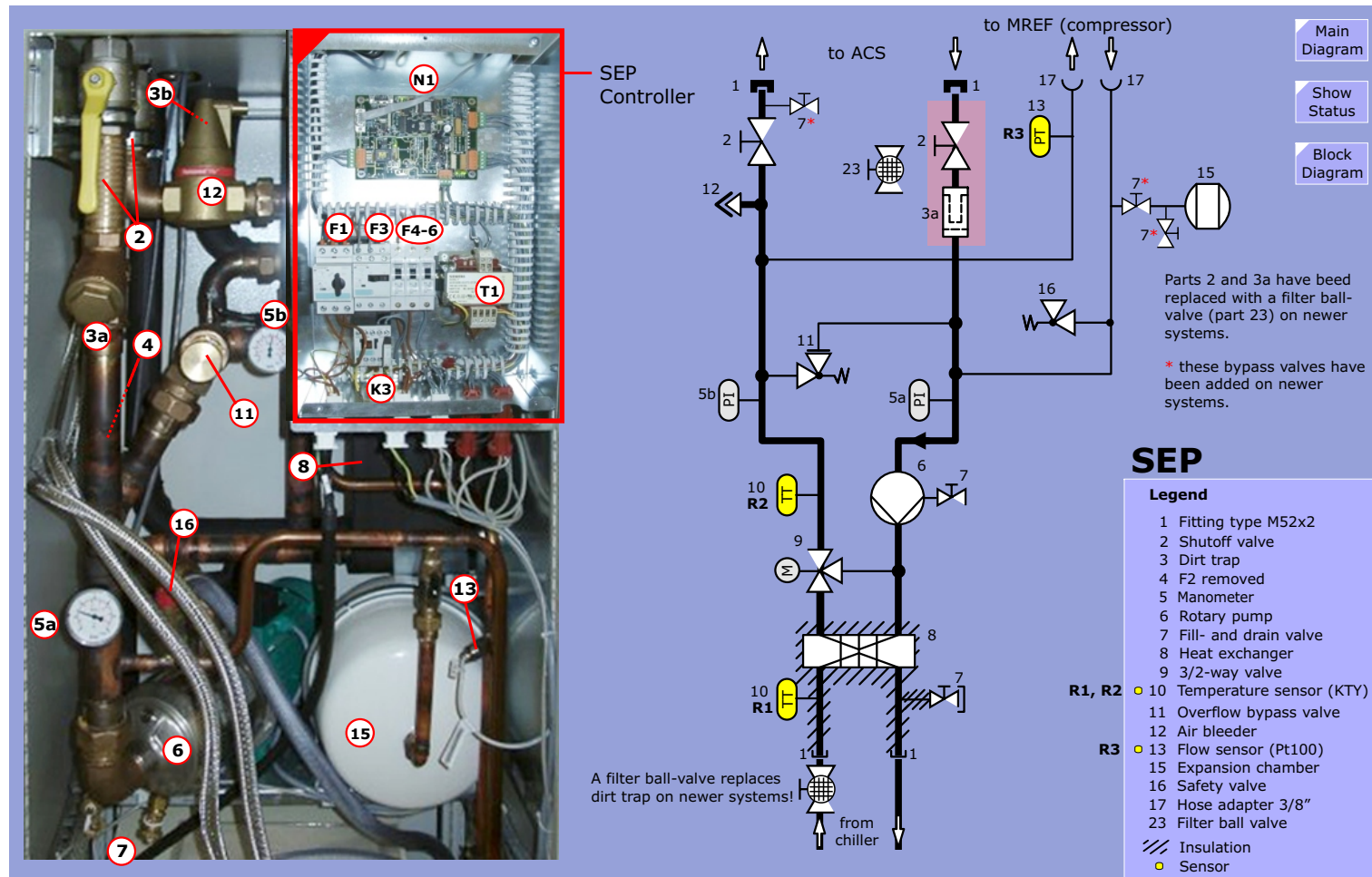
In the event the SEP were to shut off, the compressor will also be disabled. This is achieved with a switch tied into the compressor's control input (see [page 254](#)).

On systems delivered after May 2011 this circuit has been removed.

SEP Controller Parameters

Description	Type	Variable
Regulator parameters:		
Kp (P)	analog	yes
Tn (I)	analog	yes
Temp Limits:		
upper	analog	yes
lower	analog	yes
Actual value		
analog		no
Status:		
Pump on	digital	no
Service switch on	digital	no
Errors:		
Actual out of tolerance	digital	no
Dry run protection	digital	no
MREF flow	digital	no
Actual primary water temp	analog	no

Figure 196 Separator Layout



Electrical Circuit

Overview

The **Control Unit N1** reads in the analog temperature and pressure sensor values and evaluates them against limits which are sent to it via the MPCU-CAN-ACS. The limits are entered into the SeSo/Magnet and Cooling/Alarm Settings mask.

The values which are read out of the SEP controller and displayed in the Magnet and Cooling / Status page are as follows:

Status	Signal type	Value
Pump	digital	OK/Not OK
Pump overload	digital	OK/Not OK
Service switch	digital	OK/Not OK
Primary water temperature	analog	in °C
Secondary water temperature	analog	in °C
Secondary water set point	digital	in °C
MREF flow (under ACS section)	digital	OK/Not OK

Function

Circuit breaker **F1** supplies the MREF (compressor) with AC power. If it trips, a signal is sent to the controller via contact pins F1.11 and F1.14.

Circuit breaker **F3** supplies the SEP water pump. If it trips it sends a signal back to the controller via contact pins F3.11 and F3.14.

Relay **K3** is used to disable the pump if:

- the primary temperature drops below 5°C or rises above 25°C (**R1**). This will prevent pumping too cold or too warm water into the system components
- the water pressure drops below 1.5 bar (**F2**). This is too save the pump motor from burning out from a no-load situation.

In the event of a power failure, K3 will drop out turning off the Cold Head compressor. As soon as power recovers K3 will close and the compressor will automatically restart after 2 minutes.

Pressure switch **F2** signals the controller that there is insufficient water for the pump (dry run protection).

R1 is a **KTY sensor** measuring the primary water temperature. Software limits are set in the SeSo Magnet and Cooling mask and currently set to 5°C and 25°C.

R2 is a **KTY sensor** measuring the actual secondary water temperature and is used for the heat exchange regulation.

R3 is a **PT100 temp/flow sensor** indicating the water flow of the secondary. The value is read out by the MSUP and evaluated for limits.

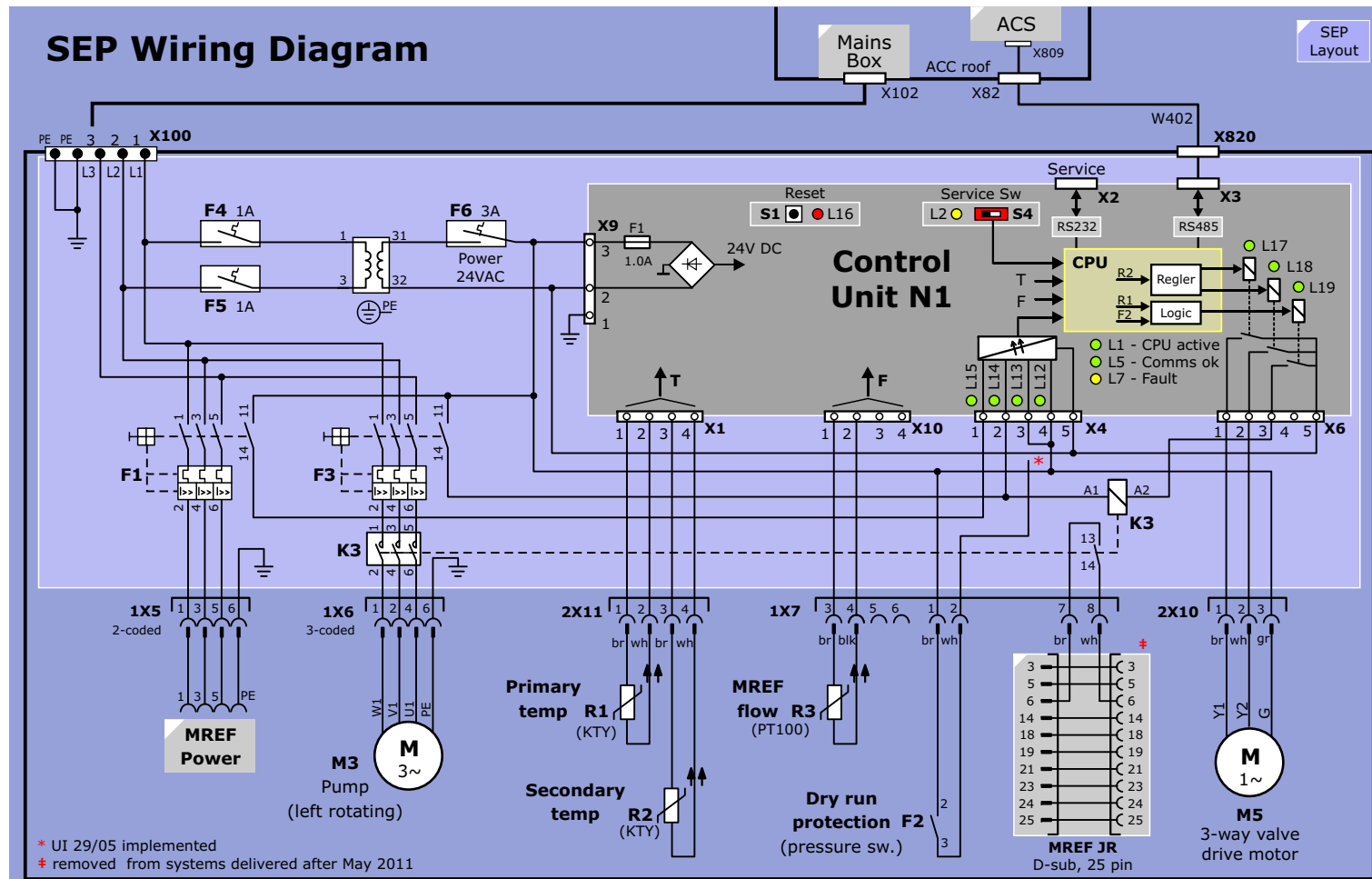
M5 drives the 3-way valve and is controlled by the Control Unit N1 via two on-board relays. The drive action is displayed by LEDs L17 (make secondary colder) and L18 (make secondary less colder).

The status is passed on to the ACS over an RS485 serial interface and from there over the CAN bus to the MPCU.

LEDs

LED	Description	normal condition
L1	CPU active	blink
L2	service mode	off
L5	comms ok	on
L7	fault	off
L12	24V AC present	on
L13	same as L12	on
L14	Pump overload (F3)	on
L15	MREF overload (F1)	on
L16	reset	off (goes on shortly when reset button S1 pressed)
L17	mix colder	on when regulating
L18	mix warmer	on when regulating
L19	pump enable (K3 enable)	on

Figure 197 Separator Wiring Diagram



PT 100 Sensor

Principle

The PT100 temperature/flow sensor is an industry standard sensor which uses platinum (thus the name PT) as the thermo-element. Its characteristic is as an PTC resistance: resistance decreases as temperature decreases. The PT100's temp/resistance characteristic is very linear and thus used as an accurate temperature sensor.

To use it as a flow sensor the thermal element is actively heated by running a constant current through it and its voltage is measured. The current heats the elements up, whereby the water flow cools it back down. The voltage is indicative of the elements resistance which is proportional to the elements temperature which in turn is proportional to the flow of water over its housing (the element is not directly in the flow of water). By setting the current the sensor can be used to measure a wide range of flows.

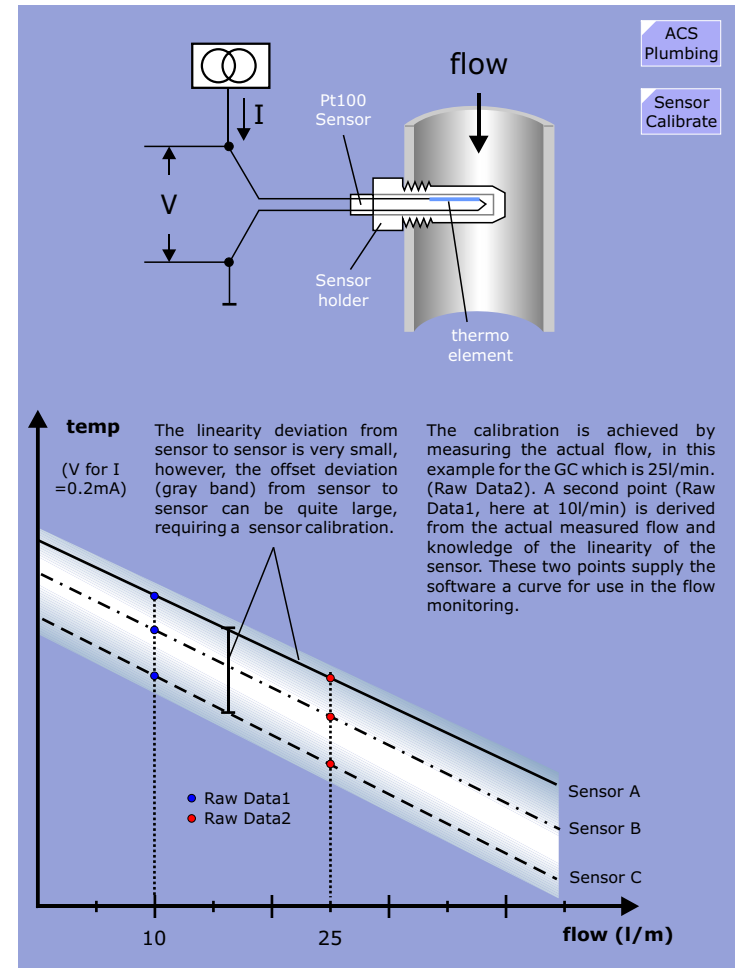
Flow Monitoring

For flow monitoring purposes, the voltage (element resistance) must be known which corresponds to the desired water flow. Although the temp/flow relationship is quite linear and does not deviate much from sensor to sensor, the actual resistance value can deviate from sensor to sensor dramatically. Therefore it is necessary to "calibrate" the sensor by reading its resistance (voltage drop) under operating conditions with the desired flow. Once the sensor resistance has been sampled the value is then used as the monitoring reference. This sensor "calibration" can be performed in the Service Software under Magnet and Cooling.

KTY Sensor

This sensor uses a semi-conductor as thermo element. Principle of operation is nearly the same as for the PT100, is not as linear however and therefore costs less.

Figure 198 PT100 Sensor



Sensor Calibration

The flow sensors must be calibrated during the system installation in order to determine the monitoring limits. During the calibration procedure, it is important that the water pump in the SEP or COCS is running and the main water flow in the case of the COCS has been set to the specifications (see Planning Guide).

NOTE If the sensor calibration is performed when there is no water flow, the water flow monitoring will never recognize a deficient water flow!

Figure 199 ACS Sensor Calibration

Calibration of Flow Sensors

- Sensor GC
- Sensor Supply (GPA, RFPA)
- Sensor Return (GC, GPA, RFPA, ACS)
- Sensor Mref

VB11 software.

Calibrate

Calibration of Flow Sensors	Raw Data1	Flow 1 l/10min	Raw Data2	Flow 2 l/10min
• Sensor GC	1263	100	1179	250
• Sensor Supply (GPA, RFPA)	1125	300	1042	700
• SensorReturn(GC, GPA, RFPA, ACS)	1090	600	1052	1200
• Sensor Mref	982	50	957	100

≥ VB13 software.

Calibrate

Monitoring

The Cooling System status masks (under SeSo > Magnet & Cooling) are given below with the corresponding sensors.

Figure 200 Cooling Status Masks

Cabinet	Air Temperature Supply	ACS R1	25.9°C
	Air Temperature Return	ACS R2	28.7°C
	Air Temperature RFIS		25.0°C
ACS	Water Temperature Cabinet		OK
	Water Pressure Supply	ACS B2	4.5bar
	Water Pressure Return	ACS B1	1.6bar
	Water Flow GC	ACS R3	OK
	Water Flow Cabinet	ACS R4	OK
	Water Flow Total	ACS R5	OK
	Water Flow Mref	IFP R1 SEP R3	OK
SEP	Pump		OK
	Pump Overload	F3	OK
	Service Switch		OK
	Water Temperature (primary)	R1	12.1°C
	Water Temperature (secondary)	R2	20.0°C
	Water (secondary) set point		20.0°C
Chiller	Chiller		OK
	Water Circuit		OK
	Refrigerant Circuit		OK
IFP	IFP		OK